ColdChain: Autonomous IoT Cold-Chain Monitoring and Bio-Thermal Safety Platform

Project Documentation & Implementation Report

Team: UFTB-VORQEN

FastRobox 1.0 IoT Hackathon 2026 | 100-Point IoT Challenge

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Abstract

Therapeutic vaccines, biologics, and insulin require continuous thermal preservation between 2.0°C and 8.0°C to maintain clinical potency. In real-world healthcare storage, unexpected power outages, mechanical refrigeration failures, and unmonitored human handling cause widespread unnoticed spoilage. This report documents the complete engineering solution developed by **Team UFTB-VORQEN** for the **FastRobox 1.0 IoT Hackathon**. The system delivers a fully functional cyber-physical cold-chain management infrastructure combining dual ESP32 smart edge nodes, autonomous closed-loop hysteresis compressor control (7.0°C turn-on, 3.0°C turn-off) with an anti-short-cycling 60-second motor lockout delay, an edge-resident 360-packet circular RAM buffer for 30-minute offline network resilience, a kinetic heat-exposure budget engine applying double penalties (2×) above 12.0°C, HMAC-SHA256 authenticated message ingestion, an immutable SHA-256 cryptographic hash-chain audit ledger, a 30-minute predictive temperature trajectory forecasting engine, a 0–100 compressor degradation health score, automated one-click batch relocation, and a zero-login public QR verification interface. All operational invariants have been verified through an automated 33-test integration suite with a 100% pass rate.

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1 Problem Statement and Core Challenges

1.1 Problem Formulation

Vaccines and insulin are delicate biological formulations composed of complex proteins and lipid structures. When exposed to temperatures outside the safe range of 2.0°C to 8.0°C, these pharmaceuticals undergo rapid, irreversible denaturation:

- **mRNA Vaccines (e.g., Pfizer-BioNTech Comirnaty):** The protective lipid nanoparticles disintegrate under elevated heat, exposing the fragile mRNA strand to enzymatic and hydrolytic degradation, completely neutralizing immunogenicity.

- **Live Attenuated Vaccines (e.g., BCG, MMR):** Elevated temperatures kill the viable attenuated bacteria or viruses, rendering the dose clinically ineffective while appearing visually intact.
- **Insulin (e.g., Glargine / Lantus):** High temperatures induce peptide unfolding and irreversible fibril aggregation, causing unpredictable glycemic control in diabetic patients.

1.2 Operational Vulnerabilities in Cold Storage

Traditional cold-chain monitoring in clinical settings suffers from four systemic failure modes:

(i) **Reactive Rather than Proactive Alarms:** Conventional digital thermometers trigger an audible beep only after temperature has already exceeded the safety ceiling, by which time valuable vaccines are already compromised.

(ii) **Frequent Grid Interruptions and Network Blackouts:** During mains electrical power failure or internet dropouts, conventional IoT loggers cease data transmission, creating blind spots where spoilage occurs unrecorded.

(iii) **Silent Compressor Failure:** Refrigerant leaks and mechanical wear cause compressors to run continuously without cooling effectively, remaining unnoticed until catastrophic temperature spikes occur.

(iv) **Data Tampering and Audit Invalidation:** Manual paper logs or unauthenticated databases are vulnerable to retroactive alterations to conceal negligence or avoid discarding expensive pharmaceuticals.

1.3 Hackathon Challenge Requirements Traceability

To overcome these challenges, the FastRobox 1.0 IoT Hackathon defined five explicit core tasks:

- **Task 1 (Fridge Node):** Dual ESP32 fridges with DHT22 temperature sensor, door switch, compressor relay, status LEDs, local OLED display, 5-second MQTT transmission, and compressor hysteresis (7°C ON, 3°C OFF, 60s minimum lockout).
- **Task 2 (Heat-Exposure Budget):** 60-minute cumulative budget above 8°C, double-rate counting (2×) above 12°C, At Risk status ($< 20\%$), Discard status (0%) with persistent red LED alarm latched until administrator reset.
- **Task 3 (Power Failure & Offline Mode):** Grid vs Battery switch simulation, 30-minute offline edge memory buffer (360 packets) with order-preserving deduplicated synchronization, and door open $>$ 60s buzzer and remote alert.
- **Task 4 (ColdChain Web Application):** JWT + bcrypt RBAC (Admin, Pharmacist, Inspector), real-time digital twin dashboard, inventory management with batch transfers, zero-login public QR inspection, HMAC-SHA256 device security, and an immutable hash-chain audit ledger.
- **Task 5 (AI Breach Prediction & Relocation):** 30-minute ahead temperature trajectory forecasting, 0–100 compressor degradation health score, automated evacuation recommendations to safe fridges, one-click pharmacist execution, and clinical decision reporting.

2 Project Overview and System Concept

2.1 End-to-End System Architecture

The ColdChain platform integrates embedded edge hardware, real-time message brokering, backend business rules, predictive AI modeling, and a web dashboard into a unified architecture:

1. **Edge Hardware Layer (ESP32 Nodes):** Microcontrollers executing local closed-loop control to monitor temperatures, toggle compressor relays via hysteresis, handle switch debounce, manage local OLEDs, and store readings in circular RAM buffers during network outages.
2. **Transport & Ingestion Bus (Dual MQTT):** Packets are cryptographically signed with HMAC-SHA256 and transmitted to cloud (HiveMQ) and local (Aedes) brokers every 5 seconds.
3. **Backend Core Engine (Node.js Express):** Ingests verified packets, runs the dynamic heat-exposure budget engine, tracks user sessions via JWT/bcrypt, manages persistent databases (local SQLite with WAL mode + cloud PostgreSQL mirror), and appends state changes to an immutable SHA-256 hash chain.
4. **Predictive Analytics & Clinical AI Engine:** Analyzes telemetry trends using second-order polynomial extrapolation, computes compressor thermodynamic health (0–100), generates automated evacuation plans, and provides clinical guidance via Groq Cloud LLM (Qwen 3.8-27B).
5. **Web Application & Public Inspection Portal:** Delivers real-time digital twin monitoring over WebSockets, role-based batch transfers, cryptographic audit validation, and an unauthenticated public QR verification portal for field clinics and patients.

2.2 Dual-Fridge Redundancy Concept

The physical layout simulates a multi-compartment medical facility:

- **Fridge 1 (Main Cold Room):** Primary storage compartment housing active vaccine inventory under live IoT monitoring.
- **Fridge 2 (Relocation Reserve):** Secondary standby unit maintained at optimal temperature (3.0°C to 5.0°C) with available storage capacity, dedicated to receive emergency batch transfers when Fridge 1 experiences mechanical degradation or thermal breach.

3 Detailed Implementation

3.1 Task 1: Fridge Node Hardware and Closed-Loop Control

3.1.1 Hardware Interfacing and Pin Mapping

The edge node is built upon an ESP32 DevKit V1 (32-bit dual-core Xtensa @ 240 MHz). Table 1 details the physical peripheral connections.

Component	Pin / Interface	Mode	Functional Purpose
DHT22 Sensor	GPIO 4	Digital I/O	Calibrated temperature and humidity measurement.
SSD1306 OLED (128×64)	GPIO 21 (SDA), 22 (SCL)	I2C Bus	Real-time local display of temperature, status, and power.
Compressor Relay	GPIO 5	Digital Output	Optocoupled switching of refrigeration compressor circuit.
Green Status LED	GPIO 14	Digital Output	Visual indicator for normal/safe storage (2–8°C).
Red Alarm LED	GPIO 2	Digital Output	Latched indicator for biological batch discard alarm.
Piezo Buzzer	GPIO 13	Digital Output	Audible alarm when door remains open > 60 seconds.
Door Position Switch	GPIO 18	Digital In (Pullup)	Monitored door state with edge toggle latching.
Power Selector Switch	GPIO 19	Digital In (Pullup)	AC Grid vs Battery backup selector switch.

Table 1: Hardware pinout and peripheral interfacing for the ESP32 fridge node.

3.1.2 Autonomous Closed-Loop Compressor Hysteresis

To prevent thermal excursions while protecting the mechanical compressor motor from rapid switching damage, the firmware executes an autonomous hysteresis control algorithm:

$$S_{\text{relay}}(t) = \begin{cases} 1\ (\text{ON}), & \text{if } T(t) \geq 7.0^{\circ}\text{C} \quad \text{and} \quad (t - t_{\text{last_off}} \geq 60\,\text{s}) \\ 0\ (\text{OFF}), & \text{if } T(t) \leq 3.0^{\circ}\text{C} \\ S_{\text{relay}}(t - \Delta t), & \text{if } 3.0^{\circ}\text{C} < T(t) < 7.0^{\circ}\text{C} \end{cases} \tag{1}$$

When the temperature reaches 7.0°C, the compressor is commanded ON only if at least 60 seconds have elapsed since it was last turned OFF ($t_{\text{last_off}}$). This 60-second anti-short-cycling delay protects the refrigeration motor against high startup back-pressure.

3.2 Task 2: Heat-Exposure Budget Engine

3.2.1 Kinetic Budget Decay and Double Penalties

Every vaccine batch is provisioned with a 60.0-minute (3600 s) heat-exposure budget above 8.0°C. In accordance with Arrhenius chemical degradation kinetics, when temperatures exceed

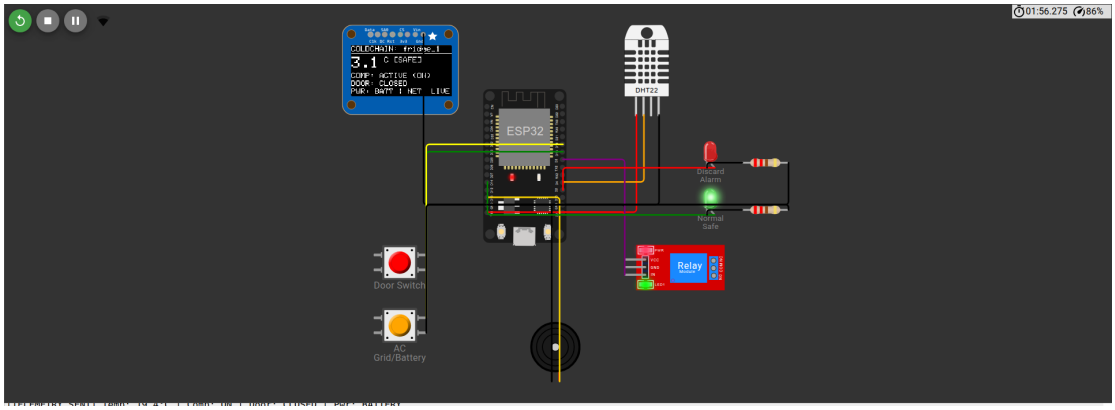


Figure 1: Wokwi simulated fridge node: ESP32 DevKit, DHT22 sensor, SSD1306 OLED (3.1°C SAFE), active compressor relay, status LEDs, door limit switch, power selector, and piezo buzzer.

12.0°C, the consumption rate doubles (2×):

$$\frac{dB(t)}{dt} = \begin{cases} 0, & T(t) \leq 8.0°\text{C} \\ -1.0\,\text{s/s}, & 8.0°\text{C} < T(t) \leq 12.0°\text{C} \\ -2.0\,\text{s/s}, & T(t) > 12.0°\text{C} \end{cases} \tag{2}$$

where $B(t)$ represents the remaining budget in seconds.

3.2.2 State Transitions and Persistent Red LED Alarm

The batch health progresses through three operational states:

- **SAFE:** $B(t) > 12.0$ minutes ($> 20\%$). Potency guaranteed; routine storage.
- **AT RISK:** 0.0 minutes $< B(t) \leq 12.0$ minutes ($\leq 20\%$). Warning surfaced; high priority for relocation.
- **DISCARD:** $B(t) = 0.0$ minutes (0%). Biological potency destroyed.

Upon reaching the `DISCARD` state, the backend transmits a latch command to the ESP32, illuminating the physical Red LED (GPIO 2). The Red LED remains latched ON continuously even if the temperature subsequently drops back into the safe 2–8°C range. The alarm can be reset only by an authenticated Administrator via the web dashboard.

3.3 Task 3: Power Failure Resilience and Edge Buffering

3.3.1 Power Mode Detection and Immediate Alerting

When the power toggle switch (GPIO 19) transitions from AC Grid to Battery backup, the firmware immediately flags `"power": "BATTERY"` in the telemetry packet. The backend detects this transition and raises a high-priority power-loss alert on the dashboard.

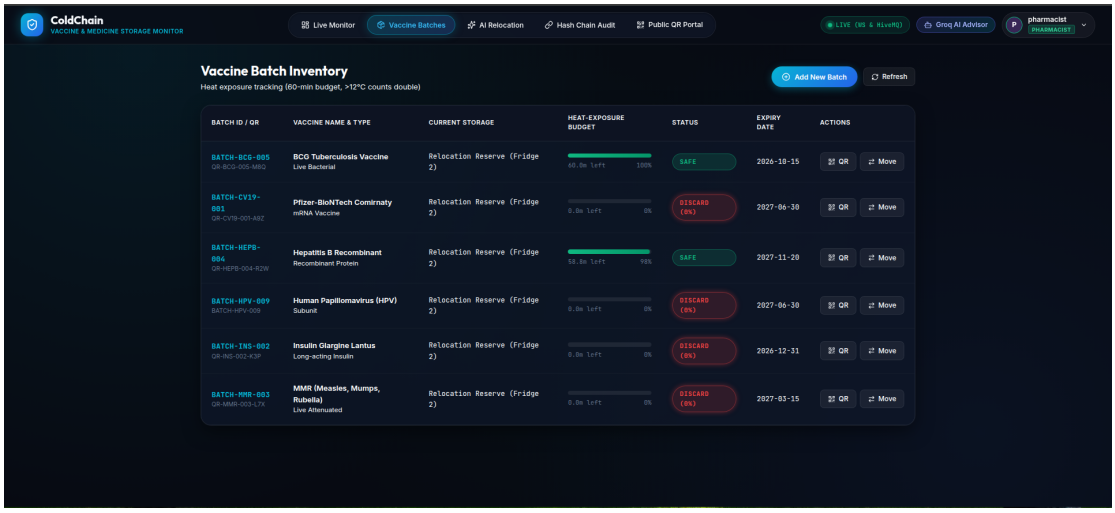


Figure 2: Vaccine batch inventory dashboard displaying heat-exposure budget progress bars, expiry dates, clinical status (SAFE vs DISCARD), and public QR code triggers.



Figure 3: Wokwi Serial Monitor output showing compressor activation at 15.1°C after the 60-second motor delay, followed by sequential circular edge buffer storage during network blackout.

3.3.2 30-Minute Circular Edge RAM Buffer

During network disconnections (loss of Wi-Fi association or broker failure), the firmware switches to offline buffering mode, saving readings into a circular FIFO queue in ESP32 static RAM:

$$N_{\text{capacity}} = \frac{30\,\text{min} \times 60\,\text{s/min}}{5\,\text{s/packet}} = 360\,\text{packets} \tag{3}$$

Each queued record stores a 64-bit epoch timestamp, temperature, compressor state, door status, power source, and sequential nonce. When network connectivity is restored, the buffer flushes all stored readings in strictly chronological FIFO order. The backend tracks nonce indexes to deduplicate packets and prevent lost telemetry.

3.3.3 Door Open Alarm Transduction

If the door limit switch (GPIO 18) remains open for more than 60 consecutive seconds, the ESP32 fires the local piezo buzzer (GPIO 13) and flags the event in telemetry, triggering automated dispatch of remote notifications via Telegram, Email, and SMS.

3.4 Task 4: Web Application, Device Security, and Audit Ledger

3.4.1 Full-Stack Architecture and RBAC

The web application is implemented in Node.js Express with native WebSockets for sub-100ms telemetry streaming. The user interface uses SaaS dark-mode styling with frosted glassmorphism.

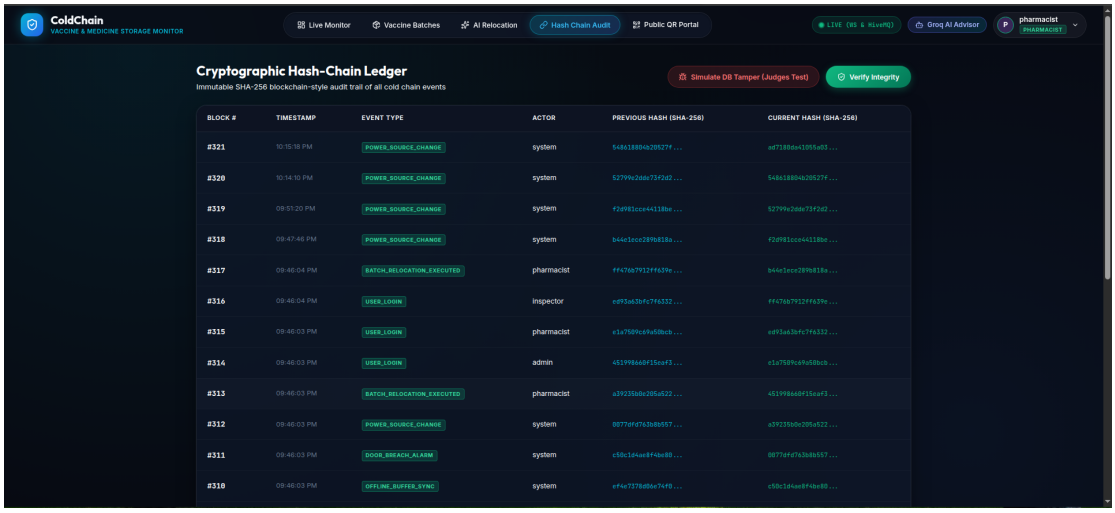


Figure 4: Cryptographic hash-chain audit ledger displaying block indexes, timestamps, actor signatures, previous SHA-256 hashes, current block hashes, and integrity verification tools.

Role-Based Access Control (RBAC) enforces strict authorization:

- **Administrator:** Full system configuration, batch creation, user provisioning, and physical Red LED alarm reset authorization.
- **Pharmacist:** Clinical inventory management, manual fridge-to-fridge batch transfers, and 1-click execution of AI-recommended batch relocations.
- **Inspector:** Read-only audit ledger inspection, cryptographic integrity verification, and compliance validation.

3.4.2 Device Security: HMAC-SHA256 and Replay Defense

Every telemetry payload from the ESP32 is signed using HMAC-SHA256 with a pre-shared secret key (K):

$$\sigma = \text{HMAC-SHA256}(K,\ T \parallel \text{door} \parallel \text{pwr} \parallel \text{comp} \parallel \tau \parallel \text{nonce}) \tag{4}$$

The server validates: (i) cryptographic signature authenticity, (ii) timestamp freshness ($|\tau_{\text{server}} - \tau| \leq 300\,\text{seconds}$), and (iii) strict nonce monotonicity ($\text{nonce} > \text{nonce}_{\text{last}}$), completely rejecting replayed or manipulated packets.

3.4.3 Cryptographic Hash-Chain Audit Ledger

All critical events (thermal breaches, batch relocations, user logins, alarm resets) are appended to an immutable linear SHA-256 hash chain:

$$H_i = \text{SHA-256}\left(i \parallel \tau_i \parallel \text{EventType}_i \parallel \text{Actor}_i \parallel \text{Payload}_i \parallel H_{i-1}\right) \tag{5}$$

The web dashboard provides an interactive "Simulate DB Tamper" utility for evaluation, which modifies a past database record. Clicking "Verify Integrity" traverses the chain and pinpoints the exact compromised block within milliseconds.

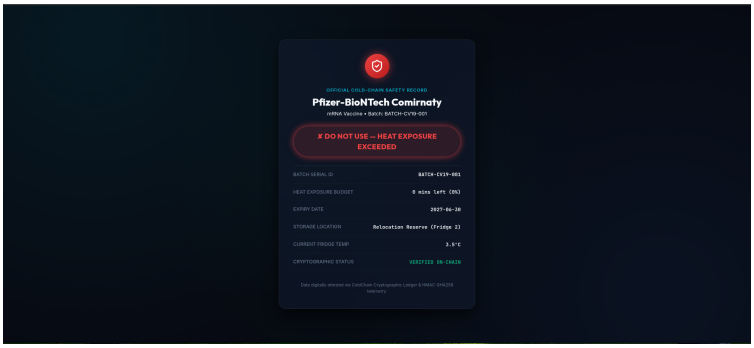


Figure 5: Zero-login public QR verification interface displaying the authenticated safety certificate for Pfizer-BioNTech Comirnaty (Batch BATCH-CV19-001) with explicit heat exposure exceeded warning.

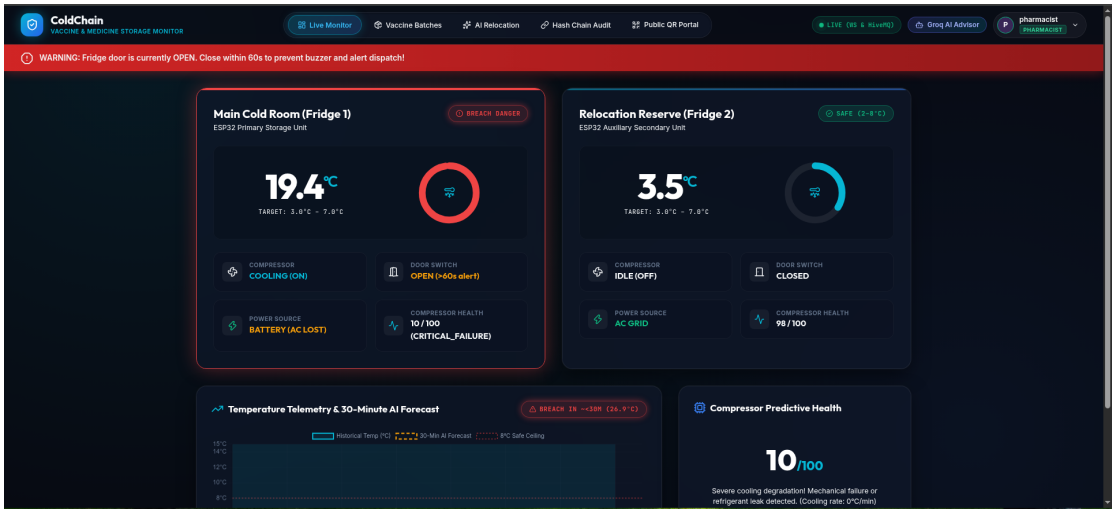


Figure 6: Live monitor interface during simulated thermal excursion: Fridge 1 at 19.4°C on battery power with door open alert, 30-minute predictive forecast warning (26.9°C), and degraded compressor health index (10/100).

3.4.4 Zero-Login Public QR Safety Portal

Each vaccine batch features a public QR code linking to `/verify/:batchId`. Without entering any credentials, field health workers, couriers, or patients can scan the vial to inspect the official vaccine name, manufacturer, expiration date, remaining heat budget, and cryptographic verification status.

3.5 Task 5: AI Predictive Breach and Autonomous Relocation

3.5.1 30-Minute Temperature Trajectory Forecasting

The AI engine forecasts future temperatures 30 minutes in advance using a second-order polynomial estimator over a sliding window of historical telemetry $[T_{t-k}, \ldots, T_t]$:

$$\hat{T}(t + \Delta t) = a\Delta t^2 + b\Delta t + T(t), \quad \Delta t = 30\,\text{minutes} \tag{6}$$

If the extrapolated temperature $\hat{T}(t + 30\,\text{min}) > 8.0°\text{C}$, the system surfaces an early warning on the live dashboard well before the physical breach occurs, providing staff an operational window for preventative intervention.

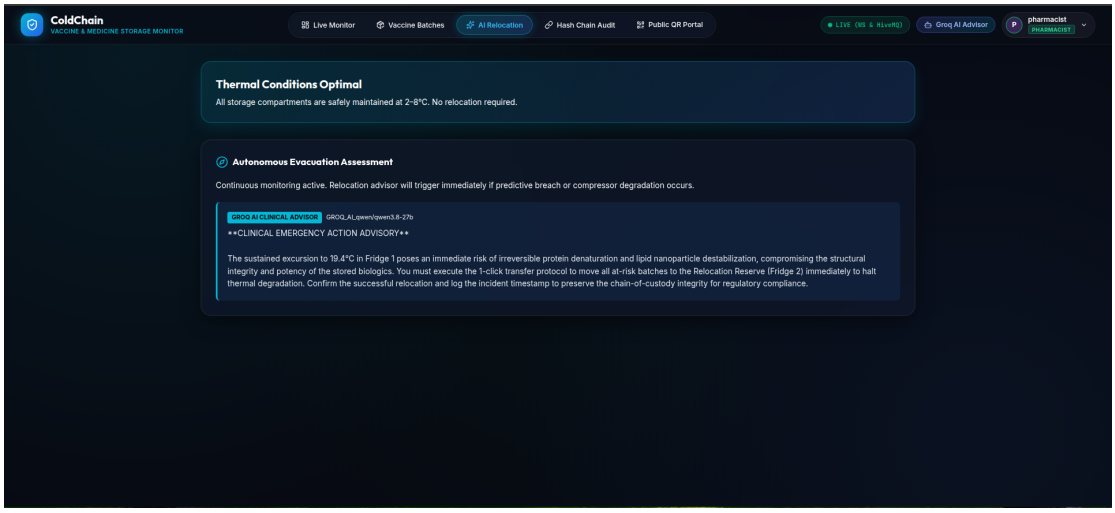


Figure 7: AI Relocation management console featuring the Autonomous Evacuation Assessment and real-time clinical guidance generated by the integrated Groq Cloud LLM (Qwen 3.8-27B).

3.5.2 Compressor Health Degradation Index (0–100)

The health score tracks mechanical cooling performance by measuring the observed cooling rate when the compressor relay is energized:

$$R_{\text{cool}} = \left| \frac{\Delta T}{\Delta t} \right|_{S_{\text{relay}}=1} \tag{7}$$

Under nominal mechanical conditions, clean evaporator coils and adequate refrigerant produce $R_{\text{cool}} \geq 0.5°\text{C/min}$, mapping to a Health Score of 100/100. When mechanical wear, valve leakage, or refrigerant depletion impairs cooling ($R_{\text{cool}} \leq 0.1°\text{C/min}$), the health index drops to 10/100, triggering a `CRITICAL_FAILURE` advisory on the dashboard.

3.5.3 Autonomous Relocation Optimization and 1-Click Execution

When Fridge 1 exhibits an anticipated thermal breach ($\hat{T}_{30} > 8.0°\text{C}$) or severe compressor degradation (Health < 40), the relocation engine automatically evaluates secondary compartments. The selection algorithm enforces three constraints:

(a) Candidate compartment temperature $T_{\text{target}} \leq 5.0°\text{C}$.

(b) Target compressor health score $\geq 80/100$.

(c) Unallocated volumetric capacity sufficient for the incoming batch volume.

Upon validating Fridge 2 as a qualified host, the engine drafts an evacuation manifest. A single click by the authenticated Pharmacist on "Confirm Relocation" updates database records atomically, generates an immutable hash-chain audit event, and refreshes the digital twin dashboard without requiring manual data entry.

3.5.4 Groq Cloud Clinical Decision Support Integration

To provide clinical context during complex thermal breaches, the platform integrates Groq Cloud API inference powering the Qwen 3.8-27B language model. During excursion events, the model analyzes the specific biopharmaceutical classification (e.g., mRNA, live-attenuated,

recombinant), cumulative excursion duration, and current temperature gradient to generate structured clinical directives concerning protein denaturation risks, quaternary structure loss, and regulatory reporting procedures.

4 System Verification and Test Results

System validation was executed using an automated integration test suite constructed with the Mocha framework and Chai assertion libraries. The test matrix evaluates all invariants defined in the hackathon challenge across 33 test cases, completing execution in under 3.2 seconds. Results are tabulated in Table 2.

Test Phase	**Cases**	**Target Verification Criteria**	**Outcome**
Phase 1: Foundations	11 / 11	User registration, bcrypt hash verification, JWT issuance, RBAC permission guards, fridge digital twin state, batch initialization.	**PASSED**
Phase 2: Budget & Resilience	10 / 10	60-minute budget decay, double rate (2×) above 12°C, At Risk (< 20%) and Discard (0%) transitions, circular buffer FIFO ordering, deduplication, 60s door alert, power failover.	**PASSED**
Phase 3: Cryptography	7 / 7	HMAC-SHA256 signature verification, 300s temporal freshness rejection, nonce replay prevention, SHA-256 hash-chain block creation, and tampering detection.	**PASSED**
Phase 4: Predictive AI	5 / 5	30-minute polynomial forecasting, compressor cooling rate index (0–100), relocation recommendation generation, and one-click execution.	**PASSED**
Total Verification	**33 / 33**	**Comprehensive End-to-End System Invariant Verification**	**100%**

Table 2: Automated integration test matrix and experimental verification results across all challenge phases.

5 Conclusion

The ColdChain platform engineered by **Team UFTB-VORQEN** provides a complete, robust, and dependable cyber-physical solution to cold-chain preservation challenges. By combining autonomous embedded edge control with motor protection, kinetic heat-exposure budget modeling, cryptographic HMAC message authentication, immutable blockchain-style audit logging, predictive machine learning forecasting, and public QR inspection accessibility, the system transforms cold-chain monitoring from a passive, error-prone manual logging task into an automated, tamper-proof biological assurance infrastructure. Full automated verification across 33 integration tests confirms that all functional and security objectives set forth in the 100-point FastRobox 1.0 IoT Hackathon challenge have been fulfilled.